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## Research Article

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# ARAFa: An LLM Generated Arabic Fact-Checking Dataset

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## Abstract

Automatic fact-checking poses a significant challenge in Arabic natural language processing due to the scarcity of datasets and resources. In this manuscript, we introduce ARAFA, a new large-scale dataset for fact-checking in Modern Standard Arabic, constructed through an automated framework leveraging large language models (LLMs). The dataset was constructed through a three-step pipeline: (1) claim generation from Arabic Wikipedia pages with supporting textual evidence, (2) claim mutation to generate challenging counterfactual claims with refuting evidence, and (3) an automatic validation step to validate that the generated claims are either supported or refuted by their accompanying evidence, or if the evidence does not provide enough information to judge the validity of the claims. The resulting dataset comprises 181,976 claim-evidence pairs labeled as *supported*, *refuted*, or *not enough information*. Human evaluation carried out on a test sample from the dataset demonstrated strong inter-annotator agreement ( $\kappa = 0.89$ ) using Cohen’s Kappa for supported claims and ( $\kappa = 0.94$ ) for refuted claims. Automatic validation based on human-evaluated sample achieved 86% accuracy for supported claims and 88% for refuted ones. To showcase ARAFA’s value as a resource for automatic Arabic fact-checking, four open-source transformer-based models were fine-tuned using ARAFA, with the top-performing model achieving a Macro F1-score of 77% on the test data. In addition to ARAFA being the first large-scale dataset for Arabic fact-checking, our framework presents a scalable approach for developing similar resources for other low-resource languages.

**Keywords:** fact-checking, Arabic NLP, claim verification, evidence retrieval

# 1 Introduction

The rapid spread of misinformation and false claims in today’s fast evolving digital landscape highlights the need for reliable automated fact-checking solutions. Automatic fact-checking has traditionally relied on deep learning models trained on fact-checking datasets [Thorne and Vlachos \(2018\)](#). Such datasets are typically generated by human annotators, which is both time consuming and resource intensive [Thorne et al. \(2018\)](#). The introduction of Large Language Models (LLMs) in recent years has paved the way for eliminating the reliance on human annotators for automatic fact-checking [Zhao et al. \(2023\)](#). Some approaches utilized LLMs to generate fact-checking datasets [Pan et al. \(2021\)](#), while others directly employed LLMs for fact-checking [Cheung and Lam \(2023\)](#). However, most of these approaches have primarily focused on high-resource languages such as English [Pan et al. \(2021\)](#); [Cheung and Lam \(2023\)](#). As a result, there remains a significant gap in fact-checking capabilities for other languages, particularly Arabic. Existing Arabic fact-checking datasets pale in comparison to their English counterparts. Datasets like AraFacts [Ali et al. \(2021\)](#) and AraStance [Alhindi et al. \(2021\)](#) are limited in scope, size and domain, primarily focusing on news or posts from X (formerly twitter), which hinders their ability to effectively train and evaluate general Arabic fact-checking models.

To address this gap, we introduce ARAFA, a large and comprehensive Arabic fact-checking dataset. Unlike existing Arabic fact-checking datasets, ARAFA is a *large general-purpose* one that can be used to train deep learning models to perform fact-checking in any domain. In addition, ARAFA can be used as a benchmark to evaluate Arabic fact-checking approaches. ARAFA consists of 181,976 natural language claims that are accompanied by textual evidence that either supports or refutes the claims, or is insufficient to judge whether the claims are true or false (i.e., not enough information, or NEI). These claims and their accompanying evidence span a wide range of topics, including History and Politics, Military and Warfare, Sports, Events and Incidents, Society and Social Issues, Science and Technology, Geography and Locations, Arts and Culture, Business and Economy, Animals and Biology, People and Biographies, Religion and Spirituality, Organizations and Institutions, as well as Environment and Nature.

In line with recent approaches that leveraged LLMs for automatic fact-checking [Pan et al. \(2021\)](#); [Cheung and Lam \(2023\)](#), ARAFA is entirely generated using LLMs, significantly alleviating the need for any manual annotation and allowing it to scale extensively. Given their demonstrated proficiency in natural language understanding, contextual reasoning, and linguistic nuance detection [Zhao et al. \(2023\)](#); [Brown et al. \(2020\)](#), LLMs have been shown to be able to generate diverse and accurate claims from reliable text sources such as Wikipedia [Pan et al. \(2021\)](#). This capability is essential for enhancing the reliability of fact-checking by ensuring that the generated claims effectively capture the complexity of natural language [Zhou et al. \(2019\)](#); [Liu et al. \(2019\)](#).

ARAFA was generated using a rigorous *fully-automated* framework, consisting of three *interleaving* tasks: a claim generation task, a claim refutation task and a validation task. In the generation task, we leverage an LLM (GPT-4o) to automate the generation of claims from Arabic Wikipedia, thereby eliminating the dependence on manual annotation. The result of this task is a set of claims along with textual evidence that supports their validity, i.e., provides enough information to verify that the claims are true based on the evidence. In the refutation task, a subset of the supported claims along with their evidence are provided to an LLM (also

**Table 1: Dataset Overview and Examples**

| Label     | Count   | Example Claim                                                                                                                          | Example Evidence                                                                                                                  |
|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Supported | 111,303 | أطلق نادي علمي في بيروت برنامج الفضاء اللبناني عام ١٩٦٠                                                                                | عام ١٩٦٠، بدأ نادي علمي من إحدى جامعات بيروت برنامجًا فضائيًا سُمي حينها برنامج الفضاء                                            |
|           |         | <i>A scientific club in Beirut launched the Lebanese space program in 1960</i>                                                         | <i>In 1960, a scientific club from one of Beirut's universities started a space program called 'Space Program'</i>                |
| Refuted   | 42,109  | يتم استخراج الألماس غالبًا من الفوهات البركانية التي تجلبه من أعماق تصل إلى ٢٠٠ كيلومترًا تحت الأرض                                    | يستخرج معظم الألماس من الفوهات البركانية حيث تلقي به الحمم البركانية التي تحضره من أعماق الأرض من مسافات قد تصل إلى ١٥٠ كيلومترًا |
|           |         | <i>Diamonds are often extracted from volcanic craters that bring them from depths up to 200 kilometers beneath the Earth's surface</i> | <i>Most diamonds are extracted from volcanic craters where lava carries them from depths up to 150 kilometers</i>                 |
| NEI       | 28,564  | إس كيو إل ألشمي يُستخدم بشكل شائع لتحديد البيانات العلائقية في البرمجيات                                                               | يمكن استخدام إس كيو إل ألشمي كمحدد للبيانات العلائقية                                                                             |
|           |         | <i>SQLAlchemy is commonly used for defining relational data in software</i>                                                            | <i>SQLAlchemy can be used as a relational data mapper</i>                                                                         |

GPT-4o) to generate counterfactual versions of the claims by mutating them using carefully-crafted techniques. The result is a set of claims that are refuted by their evidence, i.e., can be verified to be false based on the evidence. Finally, in the validation task, both supported and refuted claims and their evidence are passed to an LLM (Claude Sonnet 3.5) for validation. The validation LLM acts as a judge and decides whether each claim is indeed supported or refuted by its textual evidence, or if the evidence does not provide enough information to judge the validity of the claim, in which case the judge adjusts the claim label to NEI instead of supported or refuted. Table 1 gives an overview of the dataset, including the total number of supported, refuted and NEI claim-evidence pairs, as well as an example pair for each type.

To evaluate ARAFA, we followed a standard evaluation procedure used for many existing fact-checking datasets, which relies on human evaluation. A random sample of 200 claims generated by the claim generation task and 200 claims generated by the refutation task were annotated by two *expert* human annotators. The human annotators *collectively* labeled each claim as supported, refuted, or NEI based on its evidence. This provided ground-truth claim-evidence pairs, which were then used to assess the ability of the validation LLM in judging whether a claim is supported or refuted by its evidence, or if the evidence does not provide

enough information for either. The validation LLM achieved an accuracy of 86% with respect to the ground-truth supported claims and 88% with respect to the refuted ones. To compare this to human-level performance, the ground-truth claim-evidence pairs were also annotated by four other human annotators, *different* from the expert annotators. The annotations by the four non-expert annotators achieved strong inter-annotator agreement using Cohen’s Kappa, with  $\kappa = 0.89$  for supported claims and  $\kappa = 0.94$  for refuted claims. On the other hand, the non-expert annotators agreement with the expert ones was as high as  $\kappa = 0.86$  for supported claims, and as high as  $\kappa = 0.84$  for refuted claims. This highlights that using an LLM for validation yields comparable accuracy to human validation.

To the best of our knowledge, ARAFA is the largest Arabic fact-checking dataset, which is publicly available<sup>1</sup>. We believe that this dataset will serve as a valuable resource to advance automatic fact-checking for the Arabic language by serving as both training data and a standard benchmark. To demonstrate this, we fine-tuned four open-source transformer-based models using ARAFA, with the best-performing model AraModernBert-Base-V1.0 achieving a Macro F1-score of 77% on test data, confirming that ARAFA is indeed suitable for training a robust Arabic fact-checking approach. Moreover, our framework for generating ARAFA using LLMs presents a scalable approach for developing similar fact-checking datasets for other low-resource languages.

## 2 Related Work

Automatic fact-checking has evolved in recent years, transforming how we verify information in the digital age. Early systems used multi-stage pipelines that combined information retrieval (IR) modules with verification components [Hanselowski et al. \(2019\)](#); [Nie et al. \(2019\)](#). These systems typically retrieve evidence from external knowledge sources such as Wikipedia and then use that evidence to assess the veracity of claims. These early studies highlighted the critical role of both effective evidence retrieval and rigorous claim verification using the retrieved evidence for robust fact-checking [Hanselowski et al. \(2019\)](#); [Nie et al. \(2019\)](#). A significant milestone in the field was the introduction of the FEVER dataset [Thorne and Vlachos \(2018\)](#). FEVER, notable for its scale and breadth of topics it covers, provided claims paired with evidence sentences, categorized as *supported*, *refuted*, or *not enough information*, all derived from Wikipedia. The FEVER dataset has influenced numerous subsequent studies and research in the area of fact-checking. For instance, e-FEVER [Stammach and Ash \(2020\)](#) built upon FEVER by adding abstract summaries of the evidence, thus providing human-readable explanations of how the evidence support or refute claims. Another extension, FEVEROUS [Aly et al. \(2021\)](#), broadened the scope to include unstructured text and structured information like tables, presenting more complex and realistic fact-checking scenarios.

The field further advanced with the advent of Large Language Models (LLMs), prompting extensive research into their integration within fact-checking workflows [Zhao et al. \(2023\)](#); [Zhang and Gao \(2023\)](#); [Aly et al. \(2021\)](#). Building on this, the QACG framework was introduced [Pan et al. \(2021\)](#), which is a zero-shot fact-checking approach. QACG generates claims automatically from given evidence using question-answer pairs, reducing the need for human-annotated claims and broadening the variety of claim types fact-checking models can handle.

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<sup>1</sup><https://github.com/chriskhalil/ARAFa>

Another work [Lee et al. \(2020\)](#) explored the capabilities of pre-trained LLMs as end-to-end fact checkers, demonstrating their ability to leverage knowledge acquired during pre-training to verify facts, sometimes even without relying on external evidence sources. However, this approach also raised concerns about the transparency and potential for hallucination or fabrication of evidence by the LLM, as the source of information on which it judges the validity of claims is opaque. To this end, one approach [Liu et al. \(2019\)](#) investigated hierarchical step-by-step prompting methods to improve the explainability and performance of LLMs in news fact-checking. More recent work [Wang and Shu \(2023\)](#) looked at how LLMs can improve transparency in fact-checking by using knowledge-grounded reasoning. Finally, FactLLaMA [Cheung and Lam \(2023\)](#) explored how to optimize LLMs for fact-checking through methods like instruction tuning with external knowledge, enhancing their fact-checking abilities by grounding them in knowledge bases.

While English fact-checking has seen substantial advancements, empowered by large-scale datasets and the adaptation of LLMs, Arabic fact-checking is still in its infancy. Existing Arabic fact-checking datasets such as AraFacts [Ali et al. \(2021\)](#) and AraStance [Alhindi et al. \(2021\)](#), while valuable, are considerably small and often limited to specific domains like news or social media, representing only a fraction of the data volume available in Arabic. Other Arabic fact-checking datasets include AuRED [Haouari et al. \(2024\)](#), which focuses on Arabic rumor verification using evidence from Twitter and other sources, and AraFactEx [Althabiti et al. \(2024\)](#), which again focused on news and rumors, and was very limited in size. The scale and scope of these Arabic fact-checking datasets highlight the crucial need for larger, more diverse datasets that capture the linguistic and cultural nuances of the Arabic language, and can be used to train or evaluate robust automatic fact-checking approaches.

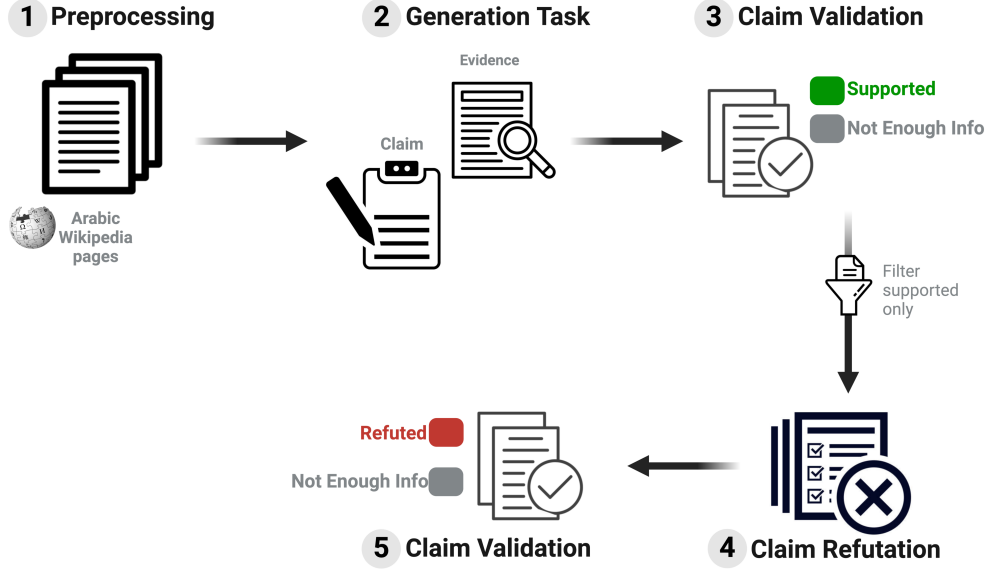
Our dataset aims to bridge this gap by being the first *publicly available* large and comprehensive (Modern Standard) Arabic fact-checking dataset. We addressed the limitations of existing Arabic dataset in terms of scale and scope by utilizing LLMs to eliminate the need for manual human annotations and to capture a diverse set of topics. While our dataset serves as a valuable resource for training and evaluating Arabic fact-checking models, our framework for the generation of the dataset can also be used to generate similar datasets for other low-resource languages with relatively simple adaptations.

### 3 Dataset Generation

ARAFA was generated using a rigorous multitask framework, which is outlined in Figure 1. The dataset generation process started by preprocessing the Arabic Wikipedia dump<sup>2</sup> to remove irrelevant content, such as category tags, internal and external links, templates, and other artifacts. Around 10,000 clean pages were then randomly sampled, covering diverse topics, including History and Politics, Military and Warfare, Sports, Events and Incidents, Society and Social Issues, Science and Technology, Geography and Locations, Arts and Culture, Business and Economy, Animals and Biology, People and Biographies, Religion and Spirituality, Organizations and Institutions, as well as Environment and Nature. The sampled pages were then chunked using a paragraph-aware chunking strategy that maintains contextual coherence, thereby mitigating the information loss typically associated with conventional text-splitting techniques that overlook natural paragraph boundaries.

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<sup>2</sup>Arabic Wikipedia Dump



**Fig. 1.** Dataset Generation Framework

Each chunk was then passed to the claim generation task, whose job was to generate a set of claims from the text chunk along with textual evidence supporting the validity of those claims using a carefully-prompted LLM. To validate that the generated claims are indeed supported by their textual evidence, the claims and their evidence were passed to another LLM that acted as a judge and checked whether each claim is indeed supported by its evidence, or whether the evidence does not provide enough information to judge the validity of the claim, in which case the claim-evidence pair was labeled NEI instead of supported. A sample of the truly supported claims were then passed to the refutation task, which applied sophisticated subtle mutations to the claims to generate counterfactual claims that are refuted by the evidence. Again, to validate that the mutated claims are refuted by their evidence, they were passed to the validation LLM, which confirmed whether they are indeed refuted or whether the evidence does not provide enough information to refute the newly generated claims. We describe each task in more details next.

### 3.1 Claim Generation Task

The goal of the generation task was to generate high-quality claim-evidence pairs from Arabic Wikipedia articles. To avoid any manual effort, and be able to scale the task to generate a large set of claim-evidence pairs, we relied on an LLM as illustrated in Figure 2. The LLM was provided with one text chunk from a Wikipedia page, and was tasked with identifying several text spans from the provided chunk to act as evidence, and then to generate a claim that is supported by each extracted evidence. We initially attempted the generation task by using a zero-shot prompt approach, which was based on the human annotation guidelines used to create previous fact-checking datasets [Thorne et al. \(2018\)](#). While this approach offered

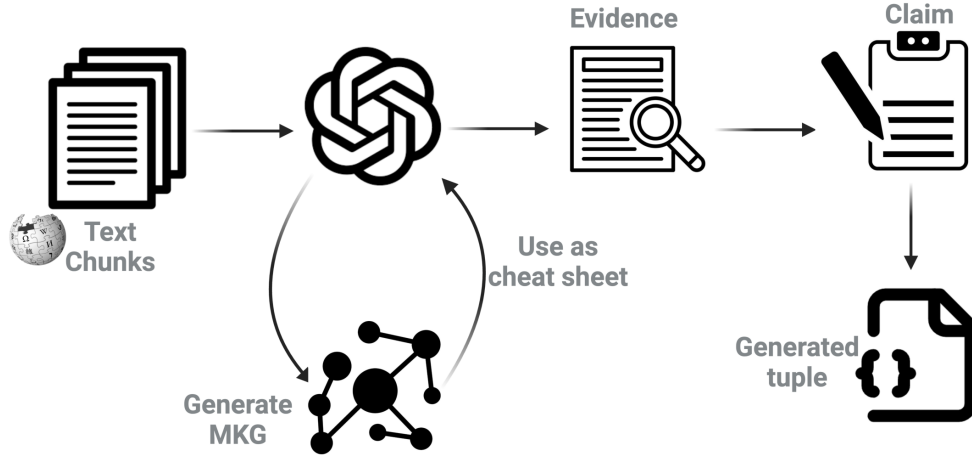


Fig. 2. Claim Generation Task

simplicity by directly integrating human annotation guidelines into a prompt, it revealed significant limitations.

First, we noticed that the LLM showed tendency to extract duplicate or near duplicate claim-evidence pairs from a given chunk, instead of creating unique pairs. Second, due to the complexity of the task, the LLM showed a high rate of hallucinations in both claim generation and evidence extraction. Instead of generating claims based on the evidence, it often paraphrased the extracted evidence to align with the claims. In other cases, the model relied on its pre-training knowledge to augment the generated claims with auxiliary information that was not present in the evidence.

A third issue with the zero-shot prompt was coreference resolution, in either the extracted evidence, the generated claims or both. Since both claims and evidence are relatively short in length, and are extracted from text chunks retrieved from Wikipedia, both the claim and the evidence need to be self contained and coreferences to named entities need to be resolved.

A fourth fundamental issue we encountered with zero-shot prompting was the poor linguistic style of the generated claims. Even though large language models are typically trained on vast, multilingual corpora, their performance can be noticeably uneven across languages [Bender et al. \(2021\)](#). In particular, these models excel in English while struggling with other languages, a disparity primarily attributed to the imbalanced nature of the training data [Arnett and Bergen \(2024\)](#); [Jung and van der Plas \(2024\)](#). For example, OpenAI’s GPT-3 was developed using a dataset composed of 93% English data and only 7% for all other languages<sup>3</sup>, underscoring the heavy bias toward English.

On the other hand, an additional challenge when particularly dealing with (Modern Standard) Arabic is its intrinsic linguistic complexity. Unlike English, which has a relatively straightforward morphology and a vocabulary size ranging from 170,000 to 600,000 words [Blog \(2023\)](#); [Abshire \(2025\)](#), Arabic features a rich morphological system yielding an estimated 12.3 million words [Abshire \(2025\)](#). For instance, Arabic adheres to strict grammatical

<sup>3</sup>GPT-3 dataset statistics

norms, particularly regarding agreement between subjects and verbs and between adjectives and nouns in terms of gender and number. Moreover, the language’s intricate system of subordinate clauses (including relative, conditional, temporal, and causal constructs) and its diverse use of coordinating conjunctions further amplify its grammatical complexity. This lexical richness not only highlights the distinctive nature of Arabic, but also necessitates specialized processing techniques to handle its complexity in generative tasks.

Recent research, such as a study by Google Research on evaluating syntactic abilities of language models, revealed that while models like BERT can learn subject–verb agreement in English, their success largely depends on the frequency of observed word patterns during training [Goldberg \(2019\)](#). Instead of comprehending grammatical principles, these models often rely on statistical correlations. This limitation is even more pronounced in Arabic, where complex morphology and a flexible word order hinder the model’s ability to generalize grammatical rules. When certain syntactic structures are underrepresented in the training data, LLMs may produce inconsistent agreement patterns. This was evident in the claims generated by the zero-shot prompt, which often had poor linguistic style and lacked grammatical agreement between nouns, adjectives and verbs when it comes to gender, number or tense.

To address all the aforementioned issues, we instead utilized a sophisticated few-shot chain-of-thought (COT) prompt, encouraging the LLM to break down the complex task of a claim generation into a series of intermediate steps, closely mimicking human-like behavior. The COT prompt instructed the LLM to first skim through the provided text chunk to get an overall understanding of the topic, then to perform an attentive reading of the chunk to identify the main named entities mentioned in the text. Next, the LLM was instructed to establish connections between the identified entities, their relationships with each other, coreferences, and any hierarchical structures. These connections formed what we refer to as the Mental Knowledge Graph (MKG), which is a mini knowledge graph generated within the model’s context window that serves as an internal "cheat sheet" to guide the claim generation process. Next, the LLM was instructed to extract multiple *verbatim* evidence texts from the given chunk, relying on the MKG, an automated similarity check, and its memory to ensure the uniqueness of the extracted evidence. To ensure that the LLM extracts evidence as it exactly appears in the chunk, it was instructed to copy the evidence as it is exactly, even if it includes spelling errors or grammatical mistakes, without any modifications or corrections.

Once evidence was extracted, the LLM was instructed to generate a complex claim from each extracted evidence that is fully supported by the evidence without external knowledge or inference, and that is distinctly different from the extracted evidence. The LLM was also instructed to ensure that the claim is semantically sound and grammatically correct by referring to a set of Arabic syntactic and semantic rules provided to it in the prompt (Arabic Rules<sup>4</sup>). To ensure coherence of the claim, the LLM was instructed to focus the claim about a single aspect of one target entity explicitly mentioned in the evidence, and to reference the target entity directly as it appears in the evidence.

To perform coreference resolution in the generated claim and its evidence, the LLM was instructed to use the MKG and the original chunk from which the evidence was extracted to map any coreferences to their named entities in the claim or the evidence. Finally, the LLM was instructed to use the MKG and the provided chunk text to do a final check as follows:

1. Verify that all the generated claims meet all the criteria for Advanced Level in Arabic.

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<sup>4</sup>Arabic Rules

2. Ensure no internal knowledge, inference or assumptions were used when generating the claims.
3. Cross-check each claim against its evidence to ensure full support and accuracy.
4. If a claim cannot be fully supported by the evidence, revise the claim or extract new evidence.
5. Ensure that each claim is not a verbatim copy or simple paraphrasing of its evidence. If it is, revise the claim to make it more complex and distinct.
6. Ensure that all the evidence extracted is unique. All evidence extracted per chunk should have a cosine similarity of 0.
7. Ensure that all the claims generated per chunk are also unique. All claims generated should have a cosine similarity below 0.80.

The full COT prompt is provided in the supplementary material. The output of this process is a set of tuples, where each tuple contains a claim, its supporting evidence, and all entity coreferences in the claim, evidence and the text chunk they were extracted from. Table 2 shows one such tuple and the original chunk from which the tuple was generated.

**Table 2:** An Example Supported Claim with Coreference Resolution

|                                                                                                                                                     |                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Original Chunk</b>                                                                                                                               |                |
| القرنة السوداء هي أعلى قمة في بلاد الشام وتقع في جبل المكمل في شمال لبنان. يبلغ ارتفاع القمة عن سطح البحر ٣٠٩٣ م.                                   |                |
| <i>Qarnat as-Sawda is the highest peak in the Levant and is located in Jabal al-Makmal in northern Lebanon. The peak is 3093 m above sea level.</i> |                |
| <b>Claim</b>                                                                                                                                        |                |
| ترتفع القرنة السوداء ٣٠٩٣ مترًا فوق سطح البحر                                                                                                       |                |
| <i>Qarnat as-Sawda rises 3093 meters above sea level</i>                                                                                            |                |
| <b>Evidence</b>                                                                                                                                     |                |
| يبلغ ارتفاع القمة عن سطح البحر ٣٠٩٣ م                                                                                                               |                |
| <i>The peak's elevation above sea level is 3093 m</i>                                                                                               |                |
| <b>Entity Coreferences</b>                                                                                                                          |                |
| Entity in Claim:                                                                                                                                    | القرنة السوداء |
| ( <i>Qarnat as-Sawda</i> )                                                                                                                          |                |
| Co-reference in Evidence:                                                                                                                           | القمة          |
| ( <i>the peak</i> )                                                                                                                                 |                |
| Co-reference in Chunk:                                                                                                                              | القرنة السوداء |
| ( <i>Qarnat as-Sawda</i> )                                                                                                                          |                |

### 3.2 Claim Refutation Task

**Table 3:** Example Mutated Claims Generated Using Different Mutation Strategies

| Strategy                     | Original Claim                                            | Mutated Claim                                             |
|------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| Temporal Nuance              | افتتحت جامعة بيروت العربية عام ١٩٦٠                       | افتتحت جامعة بيروت العربية عام ١٩٧٠                       |
|                              | <i>The Beirut Arab University was established in 1960</i> | <i>The Beirut Arab University was established in 1970</i> |
| Quantitative Distortion      | يبلغ ارتفاع جبل لبنان ٣٠٨٨ متراً                          | يبلغ ارتفاع جبل لبنان ٣٥٠٠ متراً                          |
|                              | <i>Mount Lebanon reaches a height of 3,088 meters</i>     | <i>Mount Lebanon reaches a height of 3,500 meters</i>     |
| Qualitative Shift            | تتميز بيروت بمناخها المعتدل                               | تتميز بيروت بمناخها الحار جداً                            |
|                              | <i>Beirut is characterized by its moderate climate</i>    | <i>Beirut is characterized by its very hot climate</i>    |
| Relationship Reconfiguration | طرابلس هي ثاني أكبر مدينة في لبنان                        | طرابلس هي أكبر مدينة في لبنان                             |
|                              | <i>Tripoli is the second largest city in Lebanon</i>      | <i>Tripoli is the largest city in Lebanon</i>             |
| Contextual Reframing         | يقع قصر بيت الدين في جبل لبنان                            | يقع قصر بيت الدين في شمال لبنان                           |
|                              | <i>Beiteddine Palace is located in Mount Lebanon</i>      | <i>Beiteddine Palace is located in North Lebanon</i>      |

Similar to other standard fact-checking datasets, we aimed to include false claims in ARAFA, along with evidence that clearly refutes them. To be able to do this, we sampled 50,000 supported claims that were generated using the claim generation task described in the previous subsection, and applied a refutation task to those claims to generate challenging counterfactual claims that are refuted by the evidence supporting the original claims. This is again in line with how other fact-checking datasets were constructed, albeit manually, and the intuition behind refuting supported claims rather than generating refuted claims from scratch is that it is a more challenging task (for humans and LLMs alike) to generate a false claim from a given chunk of text, and then extract a textual evidence that refutes it [Thorne et al. \(2018\)](#).

In the claim refutation task, given a supported claim tuple, which includes the original claim, its supporting evidence, and all entity coreferences, we relied on an LLM to mutate the claim in the tuple so it is refuted by the evidence while maintaining a sophisticated linguistic style similar to that of the original claim. That is, rather than simply negating statements, our refutation task employed sophisticated mutation strategies that produce natural and challenging refuted claims [Huq et al. \(2020\)](#); [Ribeiro et al. \(2018\)](#). The LLM was thus instructed

to follow carefully-crafted mutation rules to modify temporal aspects, quantitative and qualitative details, entity relationships, or contextual information, while preserving grammatical correctness and plausibility. This approach ensured that the refuted claims remained on topic and maintained semantic relevance to the original evidence. The output of the refutation task was a new tuple containing the refuted claim, along with its corresponding evidence and entity coreferences.

To highlight the importance of applying the mutation strategies when refuting claims, consider the statement: "يطل لبنان من جهة الغرب على البحر الابيض المتوسط" (Lebanon overlooks the Mediterranean Sea from the west). When the LLM was asked to refute that claim using a zero-shot prompt without providing it with the mutation strategies, it generated "يطل لبنان من جهة الشرق على البحر الاحمر" (Lebanon overlooks the Red Sea from the east). However, if the evidence makes no mention of Lebanon’s eastern border, we cannot verify that this newly generated claim is refuted by the given evidence *alone* anymore, since we do not know whether Lebanon indeed overlooks the Red Sea from the east or not just based on the evidence.

We applied the following five mutation strategies to refute claims, which were inspired by refutation tasks used in previous datasets [Thorne et al. \(2018\)](#), and strategies used for adversarial attacks on text [Huq et al. \(2020\)](#); [Ribeiro et al. \(2018\)](#), but modified to meet the particular subtleties of Arabic.

- **Temporal Nuance:** Modifies temporal aspects of a claim while maintaining the correct Arabic grammatical structure. This includes respecting Arabic’s intricate system of temporal markers and tense agreements to ensure natural linguistic flow.
- **Quantitative Distortion:** Alters numerical values or quantities while preserving the original claim’s semantic scope. This strategy carefully handles Arabic’s complex numerical system, including proper agreement rules and plural forms.
- **Qualitative Shift:** Transforms descriptive attributes of entities while adhering to Arabic’s strict adjective-noun agreement rules, ensuring consistency in gender, number, and definiteness.
- **Relationship Reconfiguration:** Modifies relationships between entities by restructuring verbal or relational constructs, while preserving the grammatical integrity of Arabic verb patterns and their syntactic implications.
- **Contextual Reframing:** Introduces broader contextual shifts so that the modified claim becomes directly refutable by the claim’s evidence.

Table 3 provides example mutated claims generated by applying the different mutation strategies outlined above.

Similar to the claim generation task, our claim refutation task utilized a thorough COT prompt, which instructed the LLM to apply the above-outlined mutation strategies to generate false claims that are (not trivially) refuted by their evidence, while at the same time making sure that the generated refuted claims are both grammatically and contextually correct. To ensure the latter, the same linguistic rules used in the prompt of the generation task were also incorporated in that of the refutation task. The complete COT prompt used in the claim refutation task is provided in the supplementary material.

### 3.3 Large Language Model Selection and Parameter Tuning

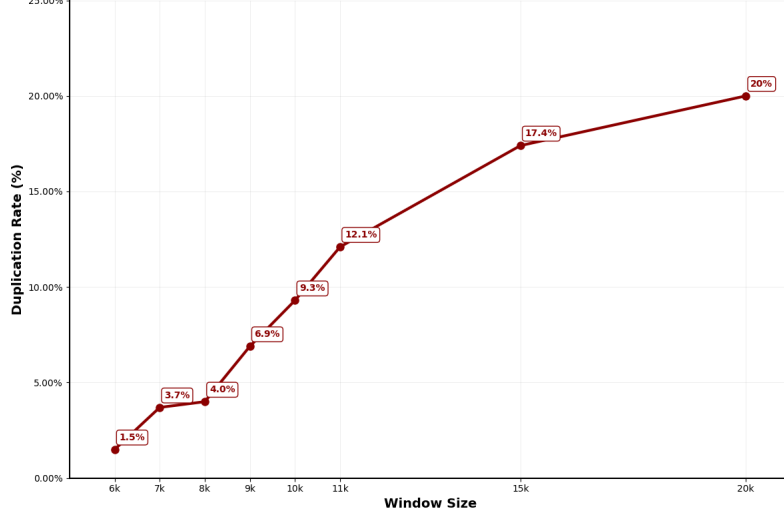
To be able to execute the claim generation and refutation tasks described above, we experimented with three major LLMs: GPT-4o, Claude Sonnet 3.5, and Llama 3.1 70B, which is an open source LLM. Note that these were considered state-of-the-art LLMs at the time the dataset was generated. Each model underwent a rigorous evaluation based on three main criteria, namely generation quality, grammatical accuracy and semantic coherence.

In the pre-selection process, we conducted an in-depth analysis of the output generated. The findings of this initial analysis showed a notable difference between the three models, especially in terms of grammatical correctness and structural coherence. Llama 3.1 70B struggled with grammatical correctness and precision despite its 70 billion parameters. The model sometimes generated out-of-context sentences and displayed recurring grammatical mistakes, despite all the efforts done with the grammar injections rules and prompt engineering. These limitations led to its disqualification before a human evaluation phase took place.

After the initial screening, the remaining two models, Claude Sonnet 3.5 and GPT-4o underwent a human evaluation. Eight annotators were presented with 20 evidence and two claims per evidence, where one claim was generated by GPT-4o and one by Claude Sonnet 3.5. The annotators were not told which claim was generated by which LLM to avoid any bias, and they were asked to choose which claim they believed was better when it comes to syntactic and semantic quality, and the ability to judge its validity based on the provided evidence.

Both Claude Sonnet 3.5 and GPT-4o generated high quality claims based on human evaluation as pointed out by the annotators. GPT-4o demonstrated greater accuracy when it comes to both the generation and refutation tasks, outperforming Claude Sonnet 3.5, which tended to be more verbose or unnecessarily complex, leading to claims that were not fully supported or fully refuted by their evidence. More precisely, the human evaluation indicated that the annotators preferred the claims generated by GPT 40% of the time compared to only 25% for Claude. They found them to be equally good 30% of the time, and they found neither claims generated by the two LLMs to be adequate only 5% of the time. For the refutation task, GPT was preferred 58.3% of the time, while Claude was preferred 33.3% of the time. Notably, the annotators never found the two models equally good (0%), and they found neither claims adequate only 8.4% of the time.

Based on the human evaluation, GPT-4o was selected as the underlying LLM for both the generation and the refutation tasks. To reduce hallucinations, *we set the model's temperature to 0*, making its responses as deterministic as possible. Finally, to determine the size of a Wikipedia page chunk that is fed to the LLM to extract evidence from, we experimented with different chunk sizes. Recall that in the generation task, the LLM was tasked with extracting multiple evidence from the same chunk, and to generate a claim for each extracted



**Fig. 3.** Rate of Duplicate Evidence Across Different Wikipedia Page Chunk Sizes

evidence. After extensive testing, we empirically determined that the optimal chunk size was 8,000 tokens. Although GPT-4o supports a 120,000-token context window, we observed that its attention mechanism suffered greatly as the chunk size increased, leading to a higher rate of duplicate evidence extraction. Specifically, using a 20,000-token chunks resulted in a 20% duplication rate, whereas reducing the chunk size to 8,000 tokens lowered this to 4%, achieving an optimal trade-off between chunk size and uniqueness as indicated in Figure 3. Similarly, the number of evidence-claim pairs generated from a given chunk was empirically set to 21.

### 3.4 Claim Validation Task

Despite the rigorous prompts that we used in the claim generation and refutation tasks, it is inevitable that the LLM would not achieve 100% accuracy on either task. This is a limitation for any automated approach that utilizes LLMs for data annotation, and in fact it is also the case when relying on human annotations [Thorne et al. \(2018\)](#). To address this, we employed an LLM-as-a-judge approach to validate that the claims generated by the generation and refutation tasks were indeed supported or refuted by their evidence. In case the validation LLM deemed an evidence to not be sufficient to fully support or fully refute a claim, it was instructed to output that the evidence does not provide enough information, which was then used to adjust the label of the claim-evidence pair from supported or refuted to NEI.

Given the concise nature of the claims in our dataset and their associated evidence, the validation task required a carefully designed approach to manage the inherent complexity of judging whether a given evidence supports, refutes a claim, or whether it does not provide enough information for either. This task can be even challenging and demanding for human judges as it requires an excessive amount of concentration, deep reasoning and careful analysis to comprehend the claim, identify any differences between it and the evidence, and make

an informed judgment. To streamline this process and make it more manageable for the validation LLM, we divided the validation task into two sub-tasks, one for supported claims and one for refuted ones, removing the need to simultaneously account for all three cases at once (i.e., supported, refuted or NEI). This is similar in spirit to breaking down a three-way classification problem into two discrete binary classification ones.

Thus, the validation LLM was first provided with the set of supported claims that were generated using our claim generation task described previously. The validation LLM was then asked to decide whether each claim is indeed supported by its evidence, or whether it needs to adjust its label to NEI. This validation task was carried out on all claim-evidence pairs that were generated using the claim generation task. Similarly, the validation LLM was provided with all the refuted claims that were generated using the claim refutation task described previously, and was asked to decide whether each claim is indeed refuted by its evidence, or whether it believes the evidence does not provide enough information to refute the claim.

Similar to the previous two tasks, the validation LLM was again instructed using a careful COT prompt (see the supplementary material for the full prompt). Our prompt was inspired by how humans perform evaluation tasks, which involves inductive reasoning and a focus on data-driven analysis that is consistent with cognitive models of human information processing [Kahneman \(2011\)](#). Our COT prompt thus mimicked, to some extent, a simplified reasoning framework of a human evaluator, thereby encouraging the LLM to process information in a more methodical and verifiable manner, leading to higher accuracy and better transparency of the output.

Building on this framework of human-inspired reasoning, the task of claim validation was broken down into simple systematic steps in which the claims were dissected into their constituent pieces and attentively validated against their evidence and entity coreferences to check for any discrepancies using predetermined criteria and a grading scheme. More precisely, the validation task involved four separate analytical steps, each methodically assessing a different facet of the claim–evidence relationship. The first step focused on the mapping between the entities mentioned in the claim and the evidence, and their coreferences. This entailed confirming the precise and unambiguous identification and mapping of the entities between the claim and the evidence. In instances where the mapping was deemed ambiguous or revealed significant discrepancies, the label of that claim-evidence pair was adjusted to NEI by the validation LLM, *and the rest of the steps were skipped*.

In the second step, the validation LLM assessed the quality of the linguistic style of both the claim and the evidence, including, but not limited to, verb-tense alignment, gender and number agreement patterns, and the consistent use of voice across statements. For example, when assessing a claim about past actions, the LLM verified that both the claim and evidence retained a compatible temporal verb form. We also emphasized the importance of the complex system of Arabic pronouns and their coreferences, ensuring that linguistic structures maintained their integrity across both the claim and the evidence. Finally, we asked the LLM to provide a quantitative linguistic score ranging between 0 and 100. This score reflected the grammatical alignment between the claim and its evidence.

In the third step, the LLM was asked to assess the vocabulary of the claim and its corresponding evidence by inspecting important terms and phrases, and their alignments. This assessment also involved matching the qualifiers across both the claim and the evidence, whereby the identical counterparts of specific terms were evaluated in both. The LLM was

**Table 4:** Validation Task Accuracy for Supported and Refuted Claims

| Claim Type    | Llama 3.1 70b | Llama 3.1 8b | GPT-4o | Claude Sonnet 3.5 | DeepSeek R1 |
|---------------|---------------|--------------|--------|-------------------|-------------|
| Supported/NEI | 71%           | 38%          | 82%    | <b>86%</b>        | 80%         |
| Refuted/NEI   | 78%           | 43%          | 85%    | <b>88%</b>        | 84%         |

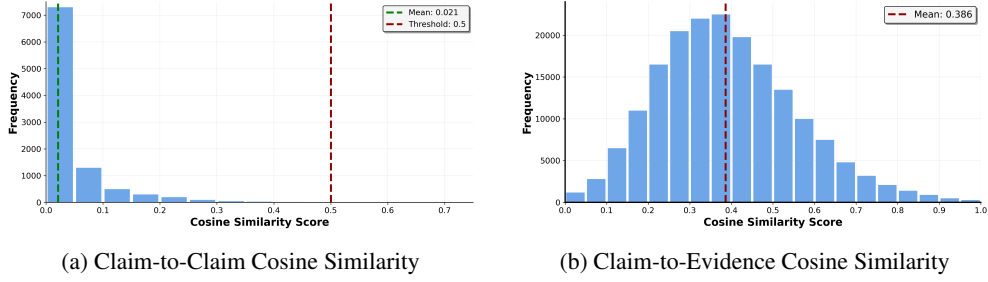
then asked to provide an alignment score ranging between 0 and 100. This score reflected the precision of terminology alignment across both the claim and the evidence.

The fourth and final step focused on semantic similarity between the evidence and the claim. The LLM was thus asked to determine semantic coherence between the claim and the evidence, which required taking scope, temporality and modality into account while meticulously examining gaps in focus, subject–predicate relations, or certainty levels. The LLM was again asked to provide a semantic similarity score that ranges between 0 and 100. Finally, these three analytical scores were aggregated into one comprehensive score that weighted the individual scores by 30%, 30%, and 40%, respectively. Any claim-evidence pair with an aggregated score less than 88 was labeled as NEI by the validation LLM.

Similar to the claim generation and refutation tasks, we experimented with various LLMs for the validation task. To be able to assess their accuracy in judging whether a claim is supported or refuted by an evidence, or whether the evidence does not provide enough information to judge the validity of the claim, we relied on human annotations to generate ground-truth claim-evidence pairs. To this end, we randomly sampled 400 claim-evidence pairs from the dataset, 200 supported claims that were generated by the claim generation task, and 200 refuted claims generated by the claim refutation task. Two (native Arabic speaker) *expert* annotators, one of whom is the lead author of this paper and another who is a PhD student, *collectively* examined each claim-evidence pair closely. The expert annotators then confirmed whether each claim is indeed supported or refuted by its evidence, or whether its label needs to be adjusted to NEI, since the evidence does not fully support or refute the claim.

Given those 400 ground-truth claim-evidence pairs, we evaluated five different LLMs in terms of their validation accuracy. Table 4 shows the accuracy of each of the five LLMs when acting as a validation LLM for both supported and refuted claims. As can be seen from the table, Claude Sonnet 3.5 led with an accuracy of 86% with respect to ground-truth *supported* claim-evidence pairs, followed by DeepSeek R1 (80%), GPT-4o (82%), and Llama 3.1 70b at approximately 71%. Similar to the case of supported claims, Claude Sonnet 3.5 had the highest accuracy among all the evaluated LLMs when validating *refuted* claims, with an accuracy of 88%, as can be seen in Table 4. Based on these results, we selected Claude Sonnet 3.5 as the final validation LLM, which was then used to validate all the claim-evidence pairs in ARAFA, and to adjust their labels to NEI if it identified any mislabeled ones.

To be able to compare the performance of the validation LLM to human-level performance, the ground-truth claim-evidence pairs were also annotated by four other human annotators, *different* from the expert annotators. All annotators were university graduates and native Arabic speakers. The non-expert annotators received direct training from the expert annotators on the validation task. The non-expert annotators were then divided into two teams (two members per team), one responsible for validating the ground-truth supported claims, and one for refuted claims. The first team was then asked to examine each claim-evidence pair, and to label it as supported if they believed the claim is supported by its evidence, or NEI



**Fig. 4.** Distribution of Cosine Similarity Scores in ARAFA

if they believed the evidence did not provide enough information to support the claim. Similarly, the second team was asked to examine each-evidence pair and to label the pair as either refuted or NEI. For the supported pairs, the inter-annotator agreement between the two non-expert annotators as measured by Cohen’s kappa was  $\kappa = 0.89$ . The first non-expert annotator achieved an agreement of  $\kappa = 0.86$  with the expert annotators, and the second annotator obtained an agreement of  $\kappa = 0.80$ . Similarly, for the refuted pairs, the inter-annotator agreement between the two non-expert annotators was  $\kappa = 0.94$ , with the first non-expert annotator achieving an agreement of  $\kappa = 0.82$  with the expert annotators, and the second achieving  $\kappa = 0.84$ . This highlights that using an LLM for validation yields comparable accuracy to human validation. On the other hand, human annotators were only able to validate an average of 60 claim-evidence pairs per hour, which clearly demonstrates the necessity of using an automated validation technique to be able to validate the generated claim-evidence pairs in ARAFA, which consists of over 180,000 claim-evidence pairs.

## 4 Dataset Analysis

ARAFA was generated using the rigorous LLM-based approach described in the previous section. It consists of 181,976 claims that are accompanied by textual evidence that either supports or refutes the claims, or is insufficient to judge whether the claims are true or false (i.e., not enough information, or NEI). The claims in ARAFA are unevenly distributed across the three classes, with the majority of the claims belonging to the *supported* class (111,303 instances), followed by *refuted* (42,109) and *NEI* (28,564). This can be attributed to various factors. First, the claim generation task is a relatively easier task since it involves extracting evidence from a Wikipedia article chunk, followed by generating a claim that is supported by the extracted evidence. The claim refutation task on the other hand is a more complex task that involves generating false claims from extracted evidence, a task that is proven to be more difficult even for humans [Thorne et al. \(2018\)](#). Finally, NEI claims were generated through the validation task, which corrected any mislabeled supported or refuted claims. Given the rigorous approaches used for both claim generation and claim refutation, only a small subset of the claim labels were adjusted to NEI through the validation task.

To demonstrate the diversity of ARAFA, we computed the similarity between its claims. Figure 4a shows a histogram of the pairwise cosine similarity scores between claims in ARAFA. As can be seen in the figure, the pairwise similarity between claims in ARAFA forms

a highly-skewed distribution with a mean cosine similarity of 0.021, indicating minimal lexical overlap between claims and suggesting the dataset captures diverse factual assertions rather than highly similar claims. Similarly, we measured the cosine similarity each claim and its evidence in ARAFA. Figure 4b shows a histogram of the cosine similarity scores between claims and their evidence. As can be seen from the figure, the similarity scores between the claims and their evidence form a more balanced distribution centered around 0.386, indicating that while claims and their evidence are lexically coherent, they are nonetheless meaningfully distinct. This confirms that the claim-evidence pairs in ARAFA are sufficiently complex to support the development of robust fact-checking approaches that move beyond basic text matching and instead require more advanced logical reasoning.

Further analysis of the claim-evidence pairs in ARAFA reveals that the claims are consistently concise with an average length of 15.41 words (median: 15.00). Similarly, the evidence in ARAFA associated with the claims is equally concise, with the majority of the evidence (89.4%) consisting of only a single sentence, and only 10.6% consisting of multiple sentences.

Finally, to demonstrate the utility of ARAFA in developing robust automatic Arabic fact-checking models, we fine-tuned four state-of-the-art transformer-based models to perform automatic fact-checking using ARAFA. All experiments were performed on a high-performance computing cluster equipped with three NVIDIA RTX A6000 GPUs, each providing 48GB VRAM for a total of 144GB GPU memory.

The first two models were encoder-based models, AraBERTv2 [Antoun et al. \(2020\)](#) and AraModernBert-Base-V1.0 [NAMAA \(2025\)](#), and were fine-tuned to perform sequence classification by concatenating claims and their evidence with "[SEP]" separator tokens as input, and the labels (*supported*, *refuted* and *NEI*) as target classes. The two models were tuned using a maximum sequence length of 512 and 1024 tokens, respectively, batch size of 32 per GPU, learning rate of  $2e-5$ , and trained for 3 epochs using the AdamW optimizer [Loshchilov and Hutter \(2017\)](#) with linear warmup.

In addition to the encoder-based models, we also fine-tuned two generative models, Llama-3.1-8B [Grattafiori et al. \(2024\)](#) and Qwen2.5-7B [Team \(2024\)](#). The generative models were fine-tuned via Parameter-Efficient Fine-Tuning (PEFT) using Low-Rank Adaptation (LoRA) [Hu et al. \(2022\)](#) with LoRA rank  $r=64$ ,  $\alpha=128$ , and dropout=0.1. The two models had a maximum sequence length of 2048 tokens, effective batch size of 33 across multiple GPUs through gradient accumulation, learning rate of  $2e-4$ , and were trained for 3 epochs using instruction-following prompts in the Llama chat format.

All four models were trained using the complete ARAFA dataset, which was split into 90% for training and 10% for validation. The models were tested using the human-annotated test set (consisting of the 400 ground-truth claim-evidence pairs) that was used for the claim validation task described in the previous section. Model selection was based on macro F1-score using the validation set and final evaluation metrics were computed using standard classification measures including per-class F1-scores and overall accuracy.

Table 5 shows the performance of the four fine-tuned models using ARAFA. As can be seen from the table, AraBERTv2 and Qwen2.5-7B achieved the highest overall accuracy (82%), while AraModernBert-Base-V1.0 exhibited the most balanced performance, attaining the highest macro F1-score (77%). Breaking down the performance by class, all four models

**Table 5:** Performance of Fine-Tuned Models Using ARAFA

| Model                   | Accuracy | F1-Macro | F1-Supported | F1-Refuted | F1-NEI |
|-------------------------|----------|----------|--------------|------------|--------|
| Arabertv2               | 82%      | 75%      | 88%          | 86%        | 51%    |
| AraModernBert-Base-V1.0 | 81%      | 77%      | 87%          | 91%        | 52%    |
| Llama-3.1-8B            | 79%      | 75%      | 47%          | 93%        | 85%    |
| Qwen2.5-7B              | 82%      | 76%      | 51%          | 88%        | 89%    |

were quite successful in detecting refuted claims with the worst model AraBERTv2, achieving an F1-score of 86% for refuted claims. On the other hand, there is a clear performance gap between encoder-based and generative models across the other two classes (supported and NEI). Encoder models (AraBERTv2, AraModernBert) excel in verifying supported claims, achieving F1-scores of 87–88%. In contrast, generative models (Llama-3.1-8B, Qwen2.5-7B) demonstrate superior capability in identifying cases where the evidence lacks sufficient information to support the claims (i.e., NEI class), with Qwen2.5-7B reaching an F1-score of 89% and Llama-3.1-8B achieving 85%, significantly outperforming encoder models (51–52%) in this category.

Notably, Llama-3.1-8B achieves exceptional performance on *refuted* claims (93% F1-score) but performs poorly on *supported* claims (47% F1-score), suggesting a conservative bias that may be influenced by our use of LoRA fine-tuning. Due to computational constraints and limited budget for training open-source large language models, we employed parameter-efficient fine-tuning using LoRA rather than full model fine-tuning. This approach, while computationally feasible, may preserve pre-trained biases and limit the model’s ability to overcome inherent skeptical tendencies, potentially explaining the asymmetric performance across the different claim classes. These complementary strengths across the different architectural paradigms strongly suggest that the use of an ensemble of different models might be the most effective approach when performing automatic fact-checking. Combining encoder models to detect true claims supported by their evidence, and generative models to determine when the evidence does not provide enough information to judge the validity of a claim could potentially yield superior overall performance, while mitigating the individual limitations observed by single models.

## 5 Conclusion

We presented ARAFA, a comprehensive and large Arabic fact-checking dataset that was fully generated using large language models (LLMs). ARAFA was generated using a rigorous pipeline consisting of three interleaved tasks, namely claim generation, claim refutation and claim validation. ARAFA consists of 181,976 claims, and where each claim is associated with textual evidence that either supports or refutes it. In cases where the evidence is not sufficient to judge the validity of a claim, the claim-evidence pair is labeled as not enough information to indicate that. The dataset was thoroughly validated using an LLM as a judge, with the judge LLM achieving an accuracy of 86% for supported claims, and 88% for refuted claims on ground-truth human-annotated claim-evidence pairs.

In addition to ARAFA serving as the largest general-purpose Arabic fact-checking dataset that can be used for training automatic fact-checking models or for evaluating them, our dataset generation framework serves as a general framework that can be used to generate similar datasets for other (low-resource) languages. ARAFA and the code to implement the framework, including the different prompts used to instruct the LLMs to perform each task is publicly available to advance automatic fact-checking efforts in Arabic and other languages. In future work, we plan to augment the dataset with claims generated from other trusted sources, different from Wikipedia. Moreover, we plan to augment ARAFA with claims and evidence in various Arabic dialects, since it currently only contains pairs in Modern Standard Arabic. Finally, we plan to continue to fine-tune small and large language models for Arabic fact-checking using ARAFA, and to compare those fine-tuned models to an approach that utilizes pre-trained LLMs directly for fact-checking, without any fine-tuning.

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- Consent for publication: All authors have provided consent for publication.
- Data availability: All data is publicly available at <https://zenodo.org/records/15020544>.
- Materials availability: All materials are publicly available at <https://github.com/chriskhalil/ARAFA>.
- Code availability: All code is publicly available at <https://github.com/chriskhalil/ARAFA>.
- Author contribution: Khalil (Conceptualization, Methodology, Implementation, Experiments, Writing), Elbassuoni (Conceptualization, Supervision, Methodology, Writing), Assaf (Conceptualization, Methodology, Supervision).

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